4.4 Saving and reusing styles

Making a good map takes time: symbols, classes, labels, diagrams… QGIS lets you save the style of a layer, so you can create and switch between different styles for the same layer. It is also possible to copy a style from one layer to another. At a broader level, you can save the current configuration of all layers together as a **map theme**.

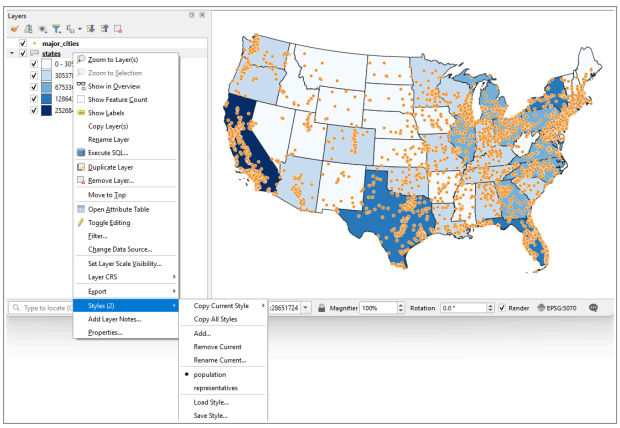
Several styles for one layer

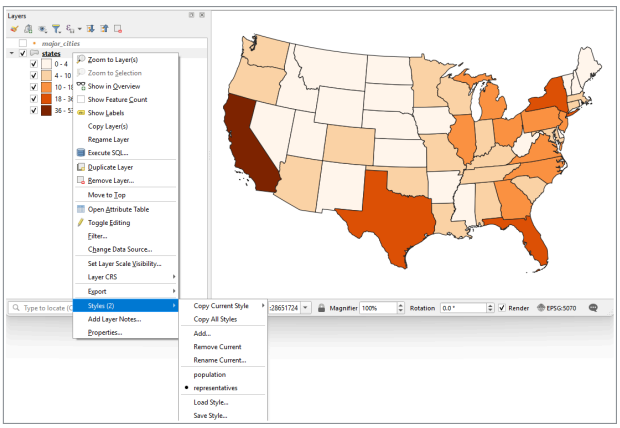
When you style a layer, you see the changes in the map canvas as soon as you apply them. This includes not only the symbology, but also the labels, the legend, diagrams and other settings. However, you may want to create different maps from the same data, each with its own styling. Rather than having to recreate the settings each time, QGIS lets you save **several named styles** for the same layer and switch between them whenever you need to.

You find the styles of a layer when you right-click it in the *Layers* panel and choose *Styles*:

- *Add…* adds a new style and gives it a name.
- The list of styles shows all styles of the layer. The active one has a dot in front of it. Click another style to switch to it.
- *Rename Current…* and *Remove Current* rename or remove the active style.

In the example below, the layer `states` has two styles: `population` (the population in 2010, left) and `representatives` (the number of representatives in 2010, right). One click in the list switches between them.





Warning

Only one style is active at a time, and every change you make is **automatically stored in the active style**. When you add a new style, QGIS switches to it immediately. The new style starts as a copy of the previous one, while the original remains unchanged. From then on, your changes are stored in the new style. This lets you experiment with a different look without losing your existing styling.

Copying a style to another layer

In the same *Styles* menu you can copy a style and paste it on another layer:

- *Copy Current Style* copies the active style. You can copy all of it, or only a part (for example only the *Symbology* or only the *Labels*).
- On the other layer, right-click and choose *Styles* ▸ *Paste Style*.

This works best when the other layer has the same geometry type and (if applicable) contains the fields the style refers to.

A style can also be saved as a file (*Save Style...*). You can then load it again (*Load Style...*), also in another project.

Map themes

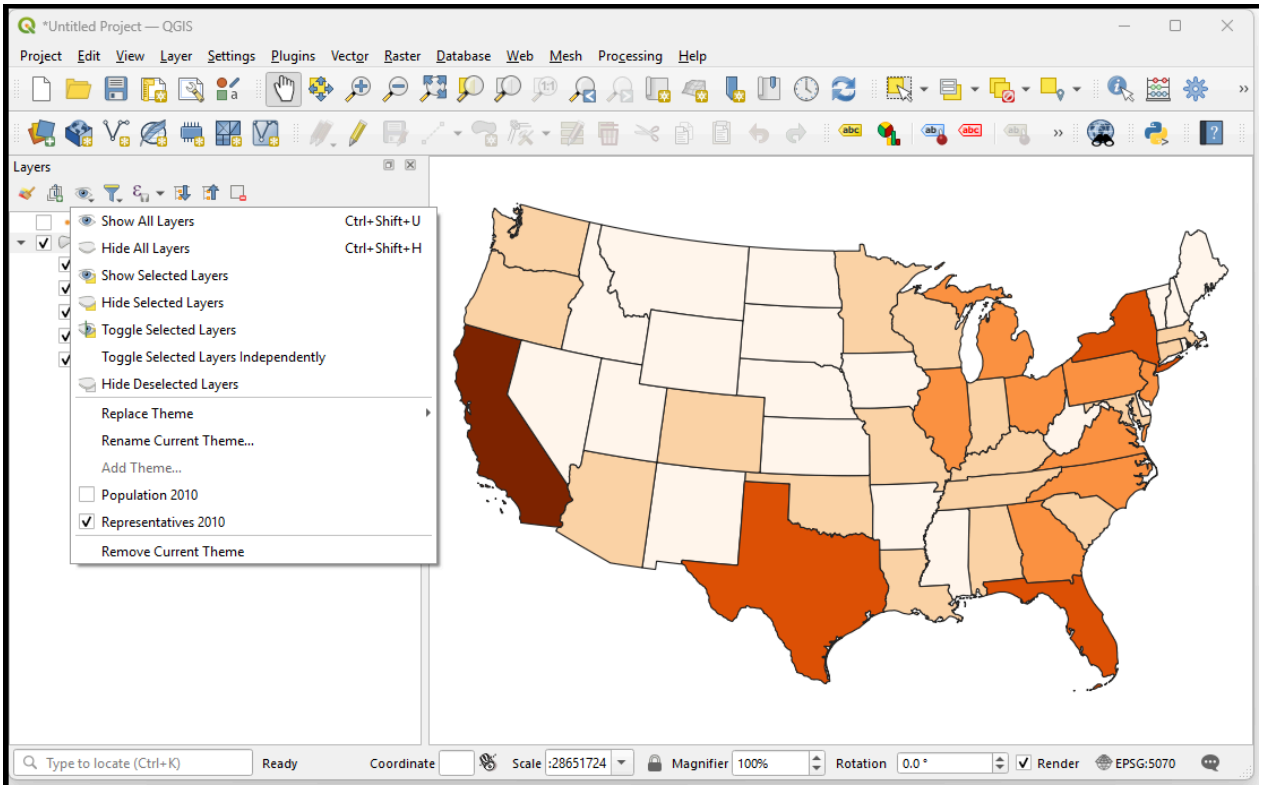
A **map theme** stores the style configuration of the whole *Layers* panel at once: for every layer, whether it is **visible**, and which of its styles is **active**. A map theme does not store the order of the layers in the *Layers* panel.



You manage the map themes with the eye icon at the top of the *Layers* panel (*Manage Map Themes*):

- *Add Theme...* saves what you see now as a new map theme.
- Click a map theme in the list to switch to it: QGIS shows and hides the layers, and activates their styles, as they were when you saved the theme.
- *Replace Theme* overwrites an existing map theme with what you see now.

In the example below there are two map themes: *Population 2010* (style `population`, with the major cities as orange dots) and *Representatives 2010* (style `representatives`, without the cities). Choosing one of them switches the style of `states` and the visibility of `major_cities` at the same time.



Warning

A map theme only stores **which** style of each layer is active, not the style itself. If you change a style after saving the theme, the map theme uses the updated style, not the style as it was when you saved the theme. A map theme also cannot bring back a layer that you removed from the project: it only works with the layers that are still in the *Layers* panel.

Conclusion

To create multiple maps with their own styles in the same QGIS project, first give the relevant layers additional styles (*Styles* ▸ *Add...*). Then save each map configuration as a separate map theme (*Manage Map Themes* ▸ *Add Theme...*). Afterwards, you can switch between the themes to display the different maps, each with its own combination of layer styles and visibility.

Warning

Make sure you don't lose your previous style: first create a new style, then start configuring.

Note

You can now do Exercise 1, parts A and B.